Better movement. Complete evidence.

Improve movement without hiding journeys, people or streets that become worse off.

Recorded research · five demand conditions

Generated demand and actual SUMO outputs. These are model experiments, not Bangalore measurements or field validation.

Protocol and provenance

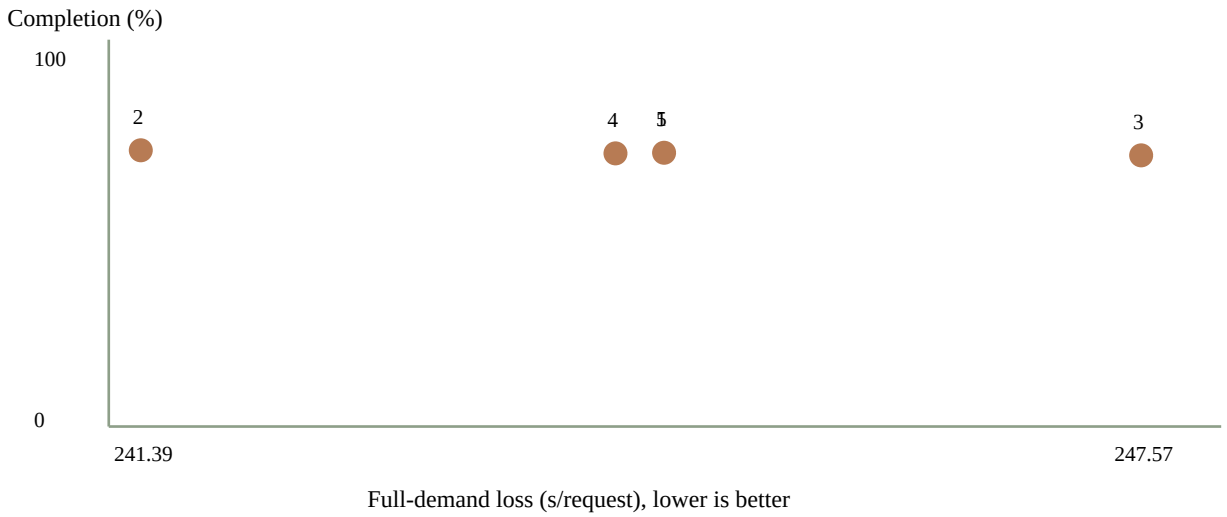
Network: synthetic. Demand: 160 requests. Departures: 120 s; clearance: 180 s. Seed sets: search [11, 12], tuning [31, 32], held-out [101, 102, 103, 104, 105, 106, 107, 108, 109, 110]. Conditions: ['uniform', 'peak', 'high_demand', 'bus_mix', 'disruption'].

Executed: 266 simulations; conservative planned maximum: 366; hard budget: 370. Candidate selection used training and tuning only. Frozen candidates were saved before evaluation.

Search: Exhaustive finite grid. Minimum completion 0.5; maximum paired completion drop 0.05; maximum invalid fraction 0.1. Constraints apply to every paired seed and condition.

Held-out outcomes

Candidate / feasible	Completion %	Loss s/request	Bus mean s	CO2 kg
Baseline / False	76.01	244.62	Unavailable	77.14
Candidate 738c / False	76.70	241.39	Unavailable	76.21
Candidate d48a / False	75.29	247.57	Unavailable	77.60
Candidate 738c · no-priority / False	75.84	244.32	Unavailable	76.32
Candidate d48a · no-priority / False	76.01	244.62	Unavailable	77.14



Numbers identify table rows. Teal: feasible; brown: constraint not met. Frontier membership is computed across the chosen objectives, not just these two axes. No claim of a global optimum.

Exact frozen interventions

Baseline: {}. Below minimum completion

Candidate 738c: {"green": 15, "transit_extension": 5}. Below minimum completion; Completion loss exceeds paired constraint

Candidate d48a: {"transit_extension": 5}. Below minimum completion; Completion loss exceeds paired constraint

Candidate 738c · no-priority: {"green": 15, "transit_extension": 0}. Below minimum completion; Completion loss exceeds paired constraint

Candidate d48a · no-priority: {"transit_extension": 0}. Below minimum completion

Paired effects by held-out condition

Candidate minus baseline. Completion is percentage points; losses and bus means are seconds. Each row uses independent seed-pair differences, not individual vehicle samples. Intervals are percentile bootstrap intervals (2,000 resamples); they are exploratory and not adjusted for multiple comparisons.

738cca548f / uniform

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	-0.71	-11.80 to 10.99	10
mean (s)	-1.73	-3.25 to -0.04	10
completion_rate (pp)	-0.12	-3.13 to 2.81	10
bus_mean (s)	-2.27	-8.42 to 3.83	10
co2_kg (kg)	-0.78	-1.84 to 0.36	10

738cca548f / peak

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	-0.16	-9.31 to 9.68	10
mean (s)	0.61	-2.26 to 3.13	10
completion_rate (pp)	0.06	-2.75 to 2.75	10
bus_mean (s)	-8.28	-12.01 to -3.50	10
co2_kg (kg)	-0.35	-1.05 to 0.59	10

738cca548f / high_demand

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	-9.01	-14.93 to -3.42	10
mean (s)	1.29	-0.15 to 2.64	10
completion_rate (pp)	2.37	0.71 to 4.12	10
bus_mean (s)	-5.16	-13.03 to 1.46	10
co2_kg (kg)	-2.08	-2.86 to -1.40	10

738cca548f / bus_mix

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	-7.75	-18.90 to 3.54	10
mean (s)	0.73	-1.47 to 3.28	10
completion_rate (pp)	2.06	-1.25 to 5.31	10
bus_mean (s)	-10.50	-16.89 to -5.14	10
co2_kg (kg)	-0.72	-1.48 to 0.05	10

738cca548f / disruption

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	1.48	-3.38 to 5.94	10
mean (s)	-3.07	-4.87 to -1.27	10
completion_rate (pp)	-0.94	-2.31 to 0.56	10
bus_mean (s)	Unavailable	Unavailable	0
co2_kg (kg)	-0.77	-1.14 to -0.46	10

d48a77c239 / uniform

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	8.08	-1.37 to 17.90	10
mean (s)	-0.40	-2.84 to 2.22	10
completion_rate (pp)	-2.38	-5.00 to 0.06	10
bus_mean (s)	9.36	6.52 to 13.25	10
co2_kg (kg)	0.51	-0.51 to 1.57	10

d48a77c239 / peak

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	4.83	0.04 to 10.25	10
mean (s)	0.82	0.03 to 1.74	10
completion_rate (pp)	-1.13	-2.56 to 0.12	10
bus_mean (s)	7.73	4.56 to 12.20	10

Outcome	Mean difference	95% interval	Seed pairs
co2_kg (kg)	0.60	0.19 to 1.12	10

d48a77c239 / high_demand

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.61	-5.13 to 6.25	10
mean (s)	1.15	-0.42 to 2.89	10
completion_rate (pp)	-0.04	-1.54 to 1.54	10
bus_mean (s)	6.98	2.07 to 12.96	10
co2_kg (kg)	0.62	-0.26 to 1.41	10

d48a77c239 / bus_mix

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	1.20	-4.35 to 7.50	10
mean (s)	1.60	-0.47 to 4.01	10
completion_rate (pp)	-0.06	-1.81 to 1.63	10
bus_mean (s)	-5.00	-9.12 to -0.93	10
co2_kg (kg)	0.52	0.06 to 1.00	10

d48a77c239 / disruption

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.00	0.00 to 0.00	10
mean (s)	0.00	0.00 to 0.00	10
completion_rate (pp)	0.00	0.00 to 0.00	10
bus_mean (s)	Unavailable	Unavailable	0
co2_kg (kg)	0.00	0.00 to 0.00	10

738cca548f-no-priority / uniform

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	4.94	-6.09 to 16.58	10
mean (s)	-2.08	-5.02 to 0.64	10
completion_rate (pp)	-1.75	-4.75 to 1.13	10
bus_mean (s)	-1.14	-5.60 to 3.08	10
co2_kg (kg)	-0.50	-1.91 to 0.83	10

738cca548f-no-priority / peak

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	9.03	0.74 to 16.87	10
mean (s)	-1.10	-2.84 to 0.48	10
completion_rate (pp)	-2.69	-5.13 to -0.19	10
bus_mean (s)	-3.41	-9.51 to 2.65	10
co2_kg (kg)	-0.06	-0.66 to 0.51	10

738cca548f-no-priority / high_demand

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	-7.31	-12.30 to -2.38	10
mean (s)	1.72	0.06 to 3.31	10
completion_rate (pp)	2.08	0.63 to 3.58	10
bus_mean (s)	-7.70	-17.96 to 1.41	10
co2_kg (kg)	-1.66	-2.56 to -0.92	10

738cca548f-no-priority / bus_mix

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	-9.64	-21.19 to 1.55	10
mean (s)	-0.15	-2.26 to 1.80	10
completion_rate (pp)	2.44	-0.81 to 5.62	10
bus_mean (s)	-10.90	-15.59 to -5.95	10

Outcome	Mean difference	95% interval	Seed pairs
co2_kg (kg)	-1.12	-1.99 to -0.31	10

738cca548f-no-priority / disruption

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	1.48	-3.38 to 5.94	10
mean (s)	-3.07	-4.87 to -1.27	10
completion_rate (pp)	-0.94	-2.31 to 0.56	10
bus_mean (s)	Unavailable	Unavailable	0
co2_kg (kg)	-0.77	-1.14 to -0.46	10

d48a77c239-no-priority / uniform

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.00	0.00 to 0.00	10
mean (s)	0.00	0.00 to 0.00	10
completion_rate (pp)	0.00	0.00 to 0.00	10
bus_mean (s)	0.00	0.00 to 0.00	10
co2_kg (kg)	0.00	0.00 to 0.00	10

d48a77c239-no-priority / peak

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.00	0.00 to 0.00	10
mean (s)	0.00	0.00 to 0.00	10
completion_rate (pp)	0.00	0.00 to 0.00	10
bus_mean (s)	0.00	0.00 to 0.00	10
co2_kg (kg)	0.00	0.00 to 0.00	10

d48a77c239-no-priority / high_demand

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.00	0.00 to 0.00	10
mean (s)	0.00	0.00 to 0.00	10
completion_rate (pp)	0.00	0.00 to 0.00	10
bus_mean (s)	0.00	0.00 to 0.00	10
co2_kg (kg)	0.00	0.00 to 0.00	10

d48a77c239-no-priority / bus_mix

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.00	0.00 to 0.00	10
mean (s)	0.00	0.00 to 0.00	10
completion_rate (pp)	0.00	0.00 to 0.00	10
bus_mean (s)	0.00	0.00 to 0.00	10
co2_kg (kg)	0.00	0.00 to 0.00	10

d48a77c239-no-priority / disruption

Outcome	Mean difference	95% interval	Seed pairs
full_demand_loss (s/request)	0.00	0.00 to 0.00	10
mean (s)	0.00	0.00 to 0.00	10
completion_rate (pp)	0.00	0.00 to 0.00	10
bus_mean (s)	Unavailable	Unavailable	0
co2_kg (kg)	0.00	0.00 to 0.00	10

Distributional impacts

Baseline

Class	Requests across runs	Completed	Loss s/request
Class · passenger	8580	6409	249.39
Class · bus	220	127	325.79

Candidate 738c

Class	Requests across runs	Completed	Loss s/request
Class · passenger	8580	6472	246.10
Class · bus	220	138	304.07

Candidate d48a

Class	Requests across runs	Completed	Loss s/request
Class · passenger	8580	6345	252.37
Class · bus	220	133	318.90

Candidate 738c · no-priority

Class	Requests across runs	Completed	Loss s/request
Class · passenger	8580	6394	249.23
Class · bus	220	145	294.55

Candidate d48a · no-priority

Class	Requests across runs	Completed	Loss s/request
Class · passenger	8580	6409	249.39
Class · bus	220	127	325.79

OD-specific sample sizes and per-run group effects remain in report.json and each run result. Groups below ten requested trips are flagged and are not ranked. Pooled group totals describe the tested runs; they are not additional statistical replicates.

Assumptions, adverse findings and reproduction

No candidate is endorsed when completion constraints fail. A reduction in emissions accompanied by fewer completed journeys remains visible. Completed-trip means are conditional on completion; unfinished travel times are censored. Full-demand loss assigns each unfinished request elapsed time since scheduled departure plus one horizon penalty. Person weighting uses scenario occupancy assumptions, not measured ridership.

Pressure control is a cyclic, bounded adaptation inspired by Varaiya (2013), The max-pressure controller for arbitrary networks of signalized intersections. It uses SUMO controlled-link movements, downstream queues and protected phase order; it does not estimate saturation flows and does not inherit theoretical stability guarantees. Vehicle physics, network

defaults and emissions classes limit interpretation.

Extract the CityLab release, install requirements.lock.txt, and run python -m scripts.build_demo with a fresh CITYLAB_DATA directory. The supplied study-spec.json, frozen.json, manifest files, immutable networks, SUMO version and application revision identify the experiment. The ZIP includes machine-readable per-seed outcomes and exact configurations. Use the in-app Evidence view for route, queue, completion, class and OD inspection.

Synthetic geometry is a constructed 4 x 4 grid. The separately included HSR OpenStreetMap extract is uncalibrated, contains disconnected components, and is attributed to OpenStreetMap contributors under ODbL. No synthetic fixture is presented as a field observation.